Macroeconomics #5

Endogenous Growth: Romer, Jones and Schumpeter

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Today's lecture

1. Introduction
2. The Romer (1990) Model
3. The Jones (1995) Model
4. The Aghion-Howitt (1992) Schumpeterian Model
5. Summary

Introduction

Why Endogenous Growth?

- Something reproducible must escape diminishing returns.
- The natural candidate is knowledge: ideas are *non-rival* (my use does not prevent yours), so a larger stock of ideas raises productivity without bound.
- But ideas are produced by people who must be paid $\Rightarrow$ some market power is needed to reward inventors.

Romer (1990): Expanding Varieties

Romer (1990): Overview

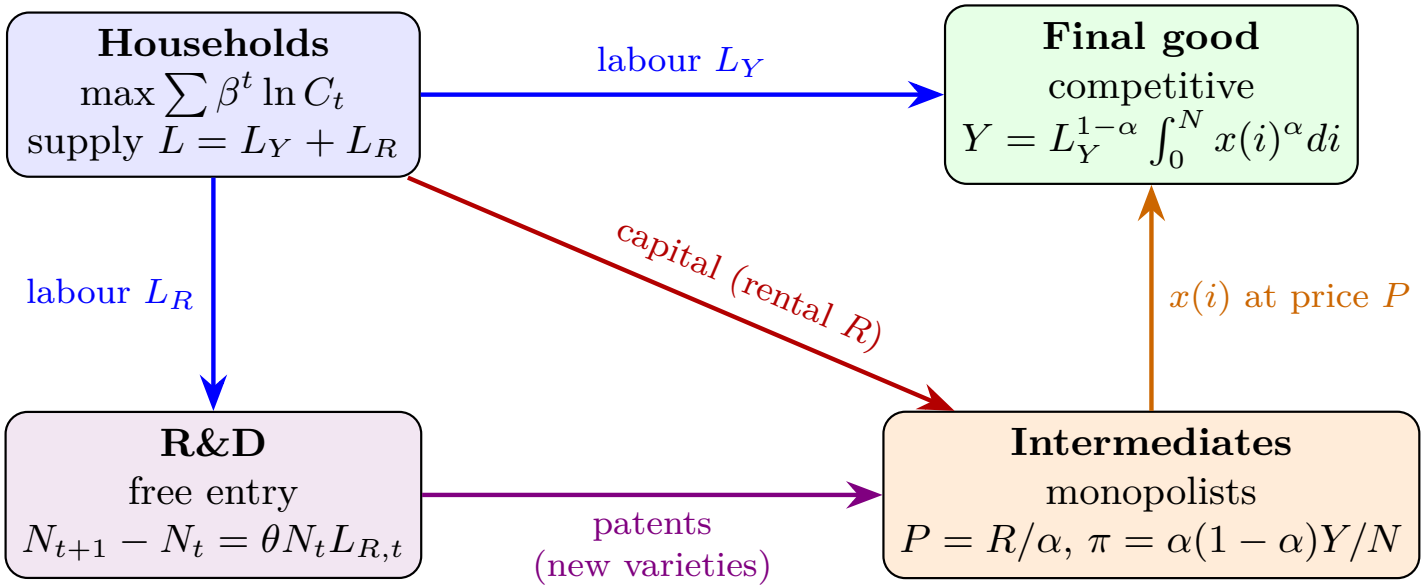
Paul Romer (1990), "Endogenous Technological Change" — Nobel Prize 2018.

- Growth is driven by **purposeful research** by profit-seeking firms.
- Entrepreneurs invent new varieties of intermediate goods (horizontal innovation).
- Each invention is protected by a **perpetual patent** ⇒ monopoly profits reward the inventor.
- More varieties ⇒ higher productivity of the final-good sector.

Four sectors, constant population L:

1. Households: consume, save, supply labour
2. Final-good sector: competitive
3. Intermediate-goods sector: monopolistic competition
4. R&D sector: free entry

Market Structure



Households own the capital and the patents (financial assets F_t).

Households

Representative household with log utility:

$$U = \sum_{t=0}^{\infty} \beta^t \ln C_t, \qquad \beta = \frac{1}{1+\rho}$$

Asset accumulation (claims on capital and patents):

$$F_{t+1} + C_t = (1 + r_t)F_t + W_t L.$$

Euler equation:

$$\boxed{\frac{C_{t+1}}{C_t} = \beta(1 + r_{t+1}) = \frac{1 + r_{t+1}}{1 + \rho}}$$

On a balanced growth path (BGP) with constant r and growth g:

$$1 + r = (1 + \rho)(1 + g) \quad \Longleftrightarrow \quad r = \rho + (1 + \rho)g \approx \rho + g.$$

Intuition: faster growth requires a higher interest rate to persuade households to postpone consumption.

Final-Good Sector

A competitive firm combines labour L_Y and a continuum of intermediates $x_t(i)$, $i \in [0, N_t]$:

$$\boxed{Y_t = L_Y^{1-\alpha} \int_0^{N_t} x_t(i)^\alpha \, di}, \qquad 0 < \alpha < 1$$

- Each variety has diminishing marginal product...
- ...but a new variety does not compete with old ones: it adds a new term to the integral.

Profit maximization gives

$$W_t = (1-\alpha)\frac{Y_t}{L_Y},$$
$$P_t(i) = \alpha L_Y^{1-\alpha} x_t(i)^{\alpha-1} \quad \text{(inverse demand)}$$

Intermediate-Goods Sector

The patent holder of variety i turns one unit of capital into one unit of x, renting capital at $R_t^k = r_t + \delta$:

$$\max_x \ \pi_t(i) = \alpha L_Y^{1-\alpha} x^\alpha - R_t^k x \quad \Longrightarrow \quad \alpha^2 L_Y^{1-\alpha} x^{\alpha-1} = R_t^k$$

$$\Longrightarrow \quad \boxed{P_t(i) = \frac{R_t^k}{\alpha}}$$

Markup

Monopoly power is the reward for invention.

Symmetric Equilibrium and Aggregation

All varieties face the same demand and cost: $x_t(i) = x_t$, $P_t(i) = P_t$, $\pi_t(i) = \pi_t$. Capital-market clearing: $N_t x_t = K_t$.

Profit per variety: since $P_t x_t = \alpha Y_t / N_t$,

$$\pi_t = (P_t - R_t^k)x_t = (1-\alpha)P_t x_t = \boxed{\alpha(1-\alpha)\frac{Y_t}{N_t}}$$

Aggregate production: substituting $x_t = K_t/N_t$,

$$Y_t = L_Y^{1-\alpha} N_t \left(\frac{K_t}{N_t}\right)^{\alpha} = \boxed{N_t^{1-\alpha} K_t^{\alpha} L_Y^{1-\alpha}}$$

- Given N: constant returns to K and L_Y.
- $N_t^{1-\alpha}$ acts as TFP: more varieties = higher productivity.
- Increasing returns to (K, L_Y, N) jointly $\Rightarrow$ sustained growth is possible.

The R&D Sector

Invention technology: researchers $L_{R,t}$ create varieties available from $t+1$:

$$\boxed{N_{t+1} - N_t = \theta N_t L_{R,t}} \quad \Longleftrightarrow \quad g_{N,t} = \theta L_{R,t}$$

A researcher's productivity θN_t rises with existing knowledge: "standing on the shoulders of giants" — the intertemporal knowledge spillover, the source of non-diminishing returns.

Value of a patent v_t (obtained at t, profits from $t+1$ on). No arbitrage with other assets:

$$1 + r_{t+1} = \frac{\pi_{t+1} + v_{t+1}}{v_t} \quad \Longleftrightarrow \quad r_{t+1} v_t = \underbrace{\pi_{t+1}}_{\text{dividend}} + \underbrace{(v_{t+1} - v_t)}_{\text{capital gain}}.$$

On the BGP Y and N grow at the same rate, so π is constant and

$$v = \pi \sum_{j=1}^{\infty} \frac{1}{(1+r)^j} = \frac{\pi}{r}.$$

Free Entry

Anybody can hire researchers. One researcher costs W_t and produces θN_t patents, each worth v_t:

$$\boxed{\theta N_t v_t = W_t}$$

- If $\theta N v > W$: unlimited entry into research.
- If $\theta N v < W$: nobody does research.

Labour market: $L_Y + L_R = L$.

BGP: L_Y, L_R constant, $g_N = \theta L_R$. From $Y = N^{1-\alpha} K^{\alpha} L_Y^{1-\alpha}$ with $g_K = g_Y$:

$$1 + g_Y = (1 + g_N)^{1-\alpha}(1 + g_Y)^{\alpha} \quad \Longrightarrow \quad g_Y = g_K = g_C = g_N = \theta L_R \equiv g.$$

Solving for Research Effort

Step 1 (free entry): $\theta N_t v = W_t = (1-\alpha)Y_t/L_Y$.

Step 2 (patent value): $v = \pi/r = \alpha(1-\alpha)(Y_t/N_t)/r$.

Step 3 (Euler on the BGP): $r = \rho + (1+\rho)\theta L_R$.

Steps 1–2: $\theta\alpha(1-\alpha)Y_t/r = (1-\alpha)Y_t/L_Y$, i.e.

$$\alpha\theta L_Y = r \quad \Longleftrightarrow \quad \alpha\theta(L - L_R) = \rho + (1+\rho)\theta L_R.$$

Solving, $\theta L_R(1+\alpha+\rho) = \alpha\theta L - \rho$:

$$\boxed{L_R = \frac{\alpha\theta L - \rho}{\theta(1+\alpha+\rho)}, \qquad g = \theta L_R = \frac{\alpha\theta L - \rho}{1+\alpha+\rho}}$$

Positive growth requires $\alpha\theta L > \rho$; otherwise $L_R = 0$.

Intuition: $\alpha\theta L_Y = r$

Free entry and $v = \pi/r$ give

$$r = \frac{\theta N \pi}{W}.$$

- Hiring one researcher costs W and creates θN patents, each paying the perpetual profit flow π.
- $\theta N \pi / W$ is the dividend yield of research; in equilibrium it equals the interest rate.
- With $\pi = \alpha(1-\alpha)Y/N$ and $W = (1-\alpha)Y/L_Y$, the yield is $\alpha\theta L_Y$: high when many workers produce, because the market for each new variety is large.

What pins down a finite growth rate?

- More research $\Rightarrow$ fewer production workers $\Rightarrow$ lower yield $\alpha\theta L_Y$.
- Faster growth $\Rightarrow$ higher r (Euler) $\Rightarrow$ lower present value of profits.

Balanced Growth: Capital, Output and Consumption

On a BGP with constant r and L_Y, the intermediate firm's FOC $\alpha^2 L_Y^{1-\alpha} x_t^{\alpha-1} = r + \delta$ gives a **constant** quantity per variety:

$$x = L_Y \left(\frac{\alpha^2}{r+\delta} \right)^{\frac{1}{1-\alpha}}.$$

- **Capital**: $K_t = N_t x \Rightarrow K_{t+1}/K_t = N_{t+1}/N_t = 1 + g$.
- **Output**: $Y_t = L_Y^{1-\alpha} N_t x^\alpha \Rightarrow Y_{t+1}/Y_t = 1 + g$; output per variety and K/Y are constant.
- **Investment**: $I_t = K_{t+1} - (1-\delta)K_t = (g+\delta)K_t$ — it equips new varieties and replaces depreciated capital.
- **Consumption**: $C_t = Y_t - I_t = \left[L_Y^{1-\alpha} x^\alpha - (g+\delta)x \right] N_t \Rightarrow C_{t+1}/C_t = 1 + g$.

$$g_C = g_Y = g_K = g_N = \theta L_R \equiv g$$

The economy grows by adding more varieties and more firms, not by making each existing firm larger: profit per patent π and the patent price $v = \pi/r$ stay constant.

How Growth is pinned down

Household (Euler) relation:

$$r = \rho + (1 + \rho)g$$

Faster consumption growth requires a higher return to persuade households to save.

Research (free-entry) relation:

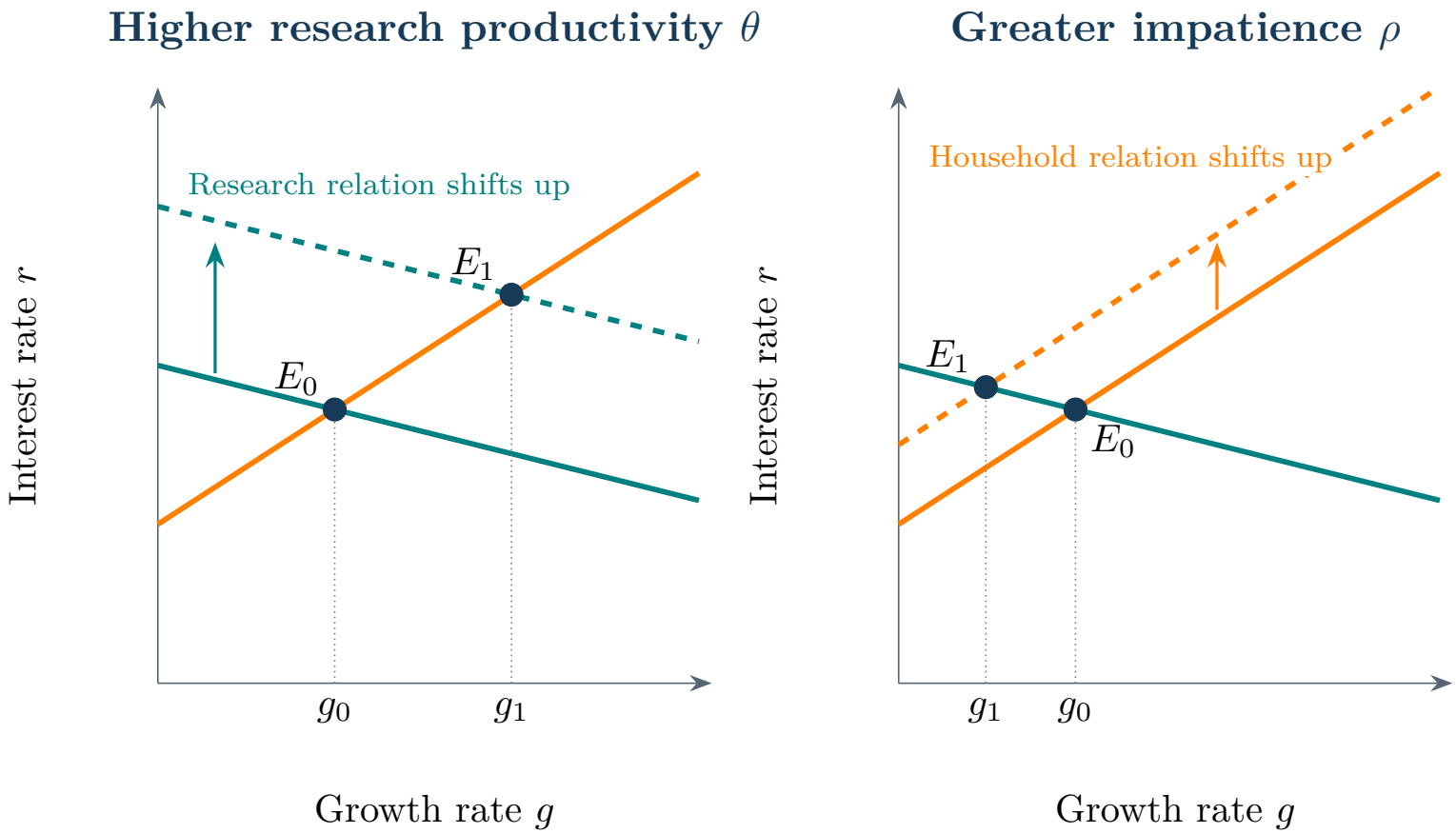
$$r = \alpha\theta L_Y = \alpha(\theta L - g)$$

More growth needs more researchers, leaving fewer production workers to generate patent profits.

The BGP is where the two lines cross: the growth rate that households are willing to finance (left) equals the growth rate that research is profitable enough to deliver (right). Solving the two equations gives

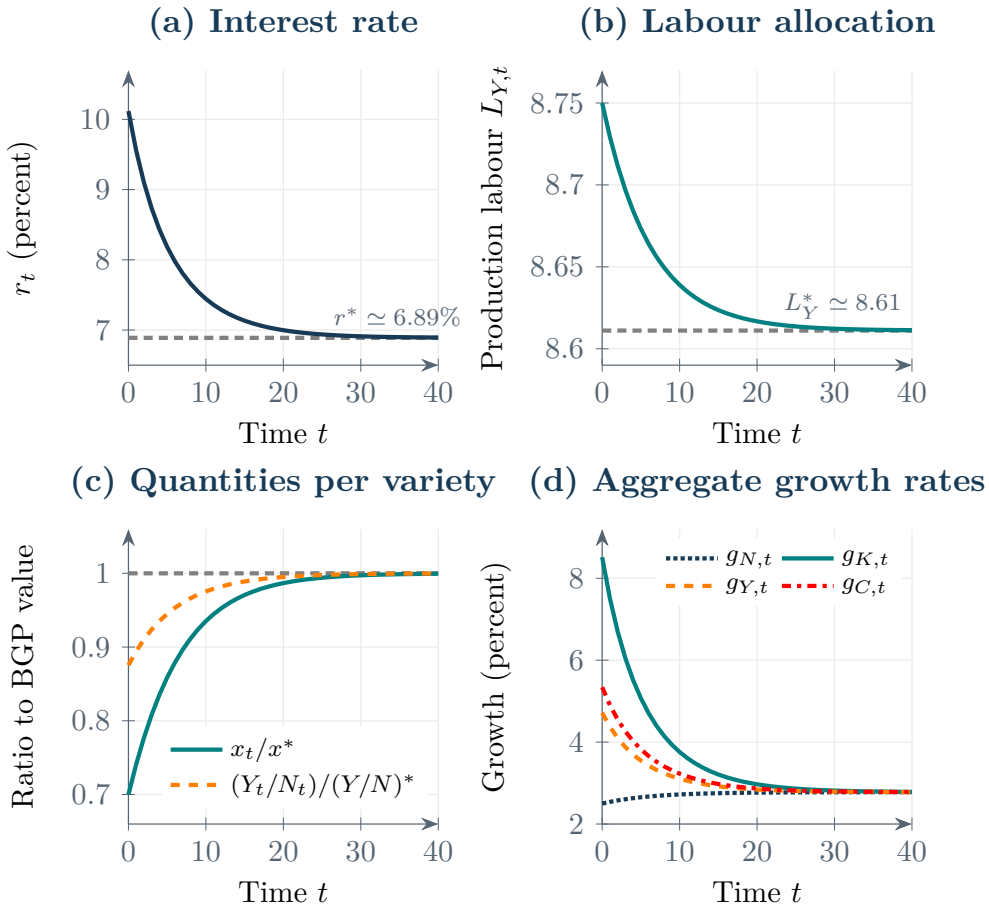
$$g = \frac{\alpha\theta L - \rho}{1 + \alpha + \rho}$$

BGP and Comparative Statics



Left: higher research productivity θ shifts the research relation up, raising g and r. Right: greater impatience ρ shifts the household relation up, lowering g and raising r.

Transition Dynamics



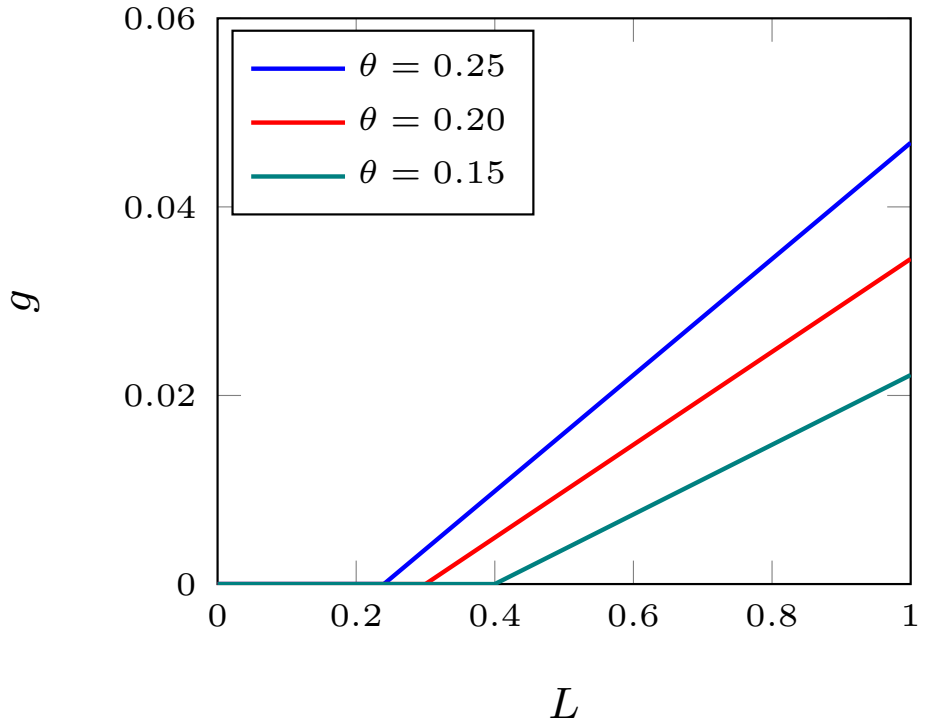
Capital initially grows faster than varieties because capital per variety is rising; output growth also reflects the changing labour allocation, and consumption growth the changing interest rate. As x_t, $L_{Y,t}$ and r_t settle at their BGP values, all four growth rates converge to the same value.

Comparative Statics and the Scale Effect

	L_R, g	Intuition
$\theta \uparrow$	$\uparrow$	research more productive
$\alpha \uparrow$	$\uparrow$	larger profit share
$\rho \uparrow$	$\downarrow$	future profits worth less
$L \uparrow$	$\uparrow$	larger market

For α: $\partial L_R/\partial\alpha$ has the sign of $L(1+\rho)+\rho/\theta > 0$.

Scale effect: g increases with the *level* of population L.



$g = \max\{0, (\alpha\theta L - \rho)/(1+\alpha+\rho)\}$, $\alpha = 1/3$,

$\rho = 0.02$.

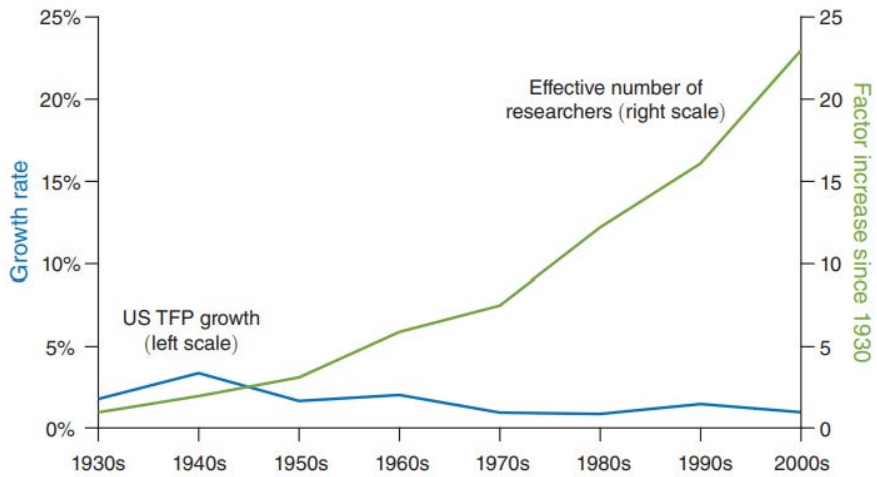
Jones (1995): Semi-Endogenous Growth

The Scale-Effect Problem

Romer's prediction: $g = (\alpha\theta L - \rho)/(1 + \alpha + \rho)$ rises with the number of people (and of researchers).

Evidence (Jones, 1995):

- The number of scientists and engineers in the US and other advanced economies has increased many-fold since the 1950s.
- Yet TFP growth and growth of income per capita have stayed roughly **constant**.



The strong scale effect is rejected by the data.

Jones (1995): Set-up

Population grows: $L_t = L_0(1+n)^t$, $n > 0$, and the R&D technology becomes

$$\boxed{N_{t+1} - N_t = \theta N_t^{\phi} L_{R,t}}, \qquad \phi < 1.$$

- $\phi < 1$: diminishing returns to existing knowledge in research — ideas get harder to find ("fishing out").
- $\phi = 1$: Romer.
- $1 - \phi$ measures how strongly the spillover is weakened.

Everything else is as in Romer.

Jones: Balanced Growth Path

$g_{N,t} = \theta N_t^{\phi-1} L_{R,t}$ is constant only if $N_t^{\phi-1} L_{R,t}$ is constant.

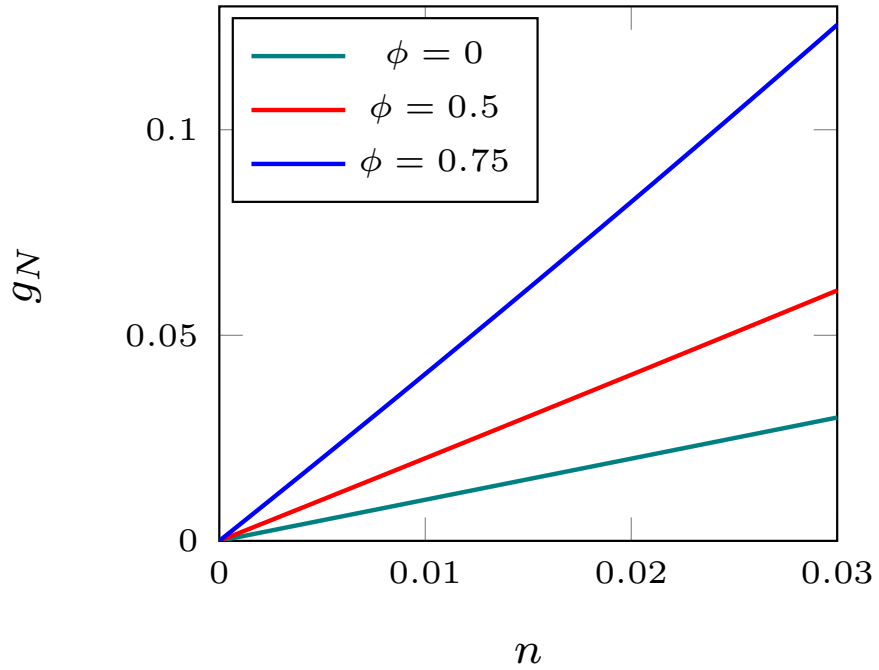
- N grows and $\phi < 1 \Rightarrow N^{\phi-1}$ falls.
- So L_R must grow, at rate n on the BGP.

Taking the ratio $g_{N,t+1}/g_{N,t}$:

$$1 = (1 + g_N)^{\phi-1}(1 + n)$$

$$\boxed{1 + g_N = (1 + n)^{\frac{1}{1-\phi}}} \qquad g_N \approx \frac{n}{1 - \phi}$$

Since $1 + g_Y = (1 + g_N)(1 + n)$, income per worker grows at g_N.



No scale effect: growth depends on n and ϕ, not on L. If $n = 0$, growth stops. Policy affects *levels*, not the long-run growth rate.

Schumpeter: Creative Destruction

Core Ideas

Joseph Schumpeter (1942): growth is a process of creative destruction — new, better products replace old ones, and the firms making the old products lose their rents.

Formalized by Aghion & Howitt (1992) and Grossman & Helpman (1991): vertical (quality-improving) innovation.

- Romer: a new variety *adds* to existing ones; patents last forever.
- Schumpeter: a new quality *replaces* the previous one; a patent is valuable only until the next innovation.

New force: business stealing — part of the innovator's reward is a transfer from the incumbent she displaces.

Environment and Timing

Constant labour $L = L_x + L_R$ (production of the intermediate, research).

Production: competitive final-good sector

$$Y_t = A_t x_t^{\alpha}, \qquad 0 < \alpha < 1,$$

A_t = quality of the leading-edge intermediate, x_t its quantity. The final-good sector also uses a fixed factor (normalized to 1) earning $(1-\alpha)Y_t$. The incumbent produces x one-for-one from labour: $x_t = L_x$.

Innovation: each researcher hired at t succeeds with probability λ, so

$$z_t = \lambda L_{R,t} \leq 1$$

is the probability of an innovation. It raises quality to γA_t ($\gamma > 1$); the winner becomes the incumbent at $t+1$; the old patent becomes worthless.

Households: risk neutral, $\beta = 1/(1+\rho) \Rightarrow$ interest rate ρ.

Prices, Wages and Profits

Inverse demand for the intermediate: $p_t = \partial Y_t / \partial x_t = \alpha A_t x_t^{\alpha-1}$.

The incumbent, with marginal cost w_t, solves

$$\max_x \ (\alpha A_t x^{\alpha-1} - w_t)x \quad \Longrightarrow \quad \alpha^2 A_t x^{\alpha-1} = w_t \quad \Longrightarrow \quad p_t = \frac{w_t}{\alpha}$$

— again a markup $1/\alpha$. In equilibrium $x_t = L_x$:

$$\boxed{w_t = \alpha^2 A_t L_x^{\alpha-1}, \qquad \pi_t = (p_t - w_t)x_t = \alpha(1-\alpha)A_t L_x^{\alpha} = \alpha(1-\alpha)Y_t}$$

- Shares: fixed factor $(1-\alpha)Y$, labour $\alpha^2 Y$, incumbent $\alpha(1-\alpha)Y$.
- Useful ratio: $\pi_t / w_t = \frac{1-\alpha}{\alpha} L_x$.

The Value of the Leading Patent

Let V_t = value at t of the patent on the current leading quality, including today's profit. With probability $1 - z$ nobody innovates and the incumbent keeps the patent:

$$V_t = \pi_t + \beta(1 - z)V_{t+1}.$$

While she survives, her quality and profit stay at $\pi(A_t)$, so

$$V(A) = \frac{\pi(A)}{1 - \beta(1 - z)} = \pi(A)\frac{1 + \rho}{\rho + z}.$$

Creative destruction as an extra discount rate

Define $1 + \rho_{\text{eff}} = 1/[\beta(1 - z)]$:

$$\rho_{\text{eff}} = \frac{\rho + z}{1 - z} \approx \rho + z.$$

A higher innovation rate shortens the expected life of a patent and lowers its value. E.g. $\rho = 2\%$: $z = 2\%$ gives 4.1%; $z = 50\%$ gives 104%.

Research Arbitrage

A research firm hires L_R researchers at t. With probability λL_R it obtains the patent on γA_t, exploited from $t+1$; its value today is $\beta V(\gamma A_t) = \beta\gamma V(A_t)$ (value is proportional to quality).

Free entry (zero expected profit):

$$\boxed{w_t = \beta\lambda\gamma V(A_t) = \lambda\gamma \frac{\pi(A_t)}{\rho + z}}$$

- LHS: what a worker earns for sure in production.
- RHS: expected value of what she produces in research.

Equilibrium Innovation

Substitute w_t and π_t (the A_t cancels):

$$\alpha^2 A_t L_x^{\alpha-1} = \lambda\gamma \frac{\alpha(1-\alpha)A_t L_x^{\alpha}}{\rho + z} \quad \Longrightarrow \quad \alpha(\rho + z) = \lambda\gamma(1-\alpha)L_x.$$

Labour market: $z = \lambda(L - L_x)$. Solving:

$$\boxed{L_x = \frac{\alpha(\rho + \lambda L)}{\lambda\big[\alpha + \gamma(1-\alpha)\big]}, \qquad z = \lambda(L - L_x)}$$

(parameters such that $0 < L_x < L$ and $z \leq 1$).

Worked example: $\alpha = 1/2$, $\gamma = 1.05$, $\lambda = 1$, $L = 1$, $\rho = 0.02$:

- $L_x = 0.51/1.025 \approx 0.50$, $z \approx 0.50$
- $\pi = 0.25 \times 0.4976^{0.5} \approx 0.176$, $V = 0.176 \times 1.02/0.5224 \approx 0.34$
- $\rho_{\text{eff}} \approx 105\%$: a very short-lived patent

Growth

$Y_t = A_t L_x^{\alpha}$ with L_x constant, so output grows with quality:

$$\frac{Y_{t+1}}{Y_t} = \begin{cases} \gamma & \text{with probability } z, \\ 1 & \text{with probability } 1-z, \end{cases} \qquad \mathbb{E}\Big[\ln \frac{Y_{t+1}}{Y_t}\Big] = \boxed{z \ln \gamma}$$

and $\mathbb{E}[Y_{t+1}/Y_t] = 1 + z(\gamma - 1)$.

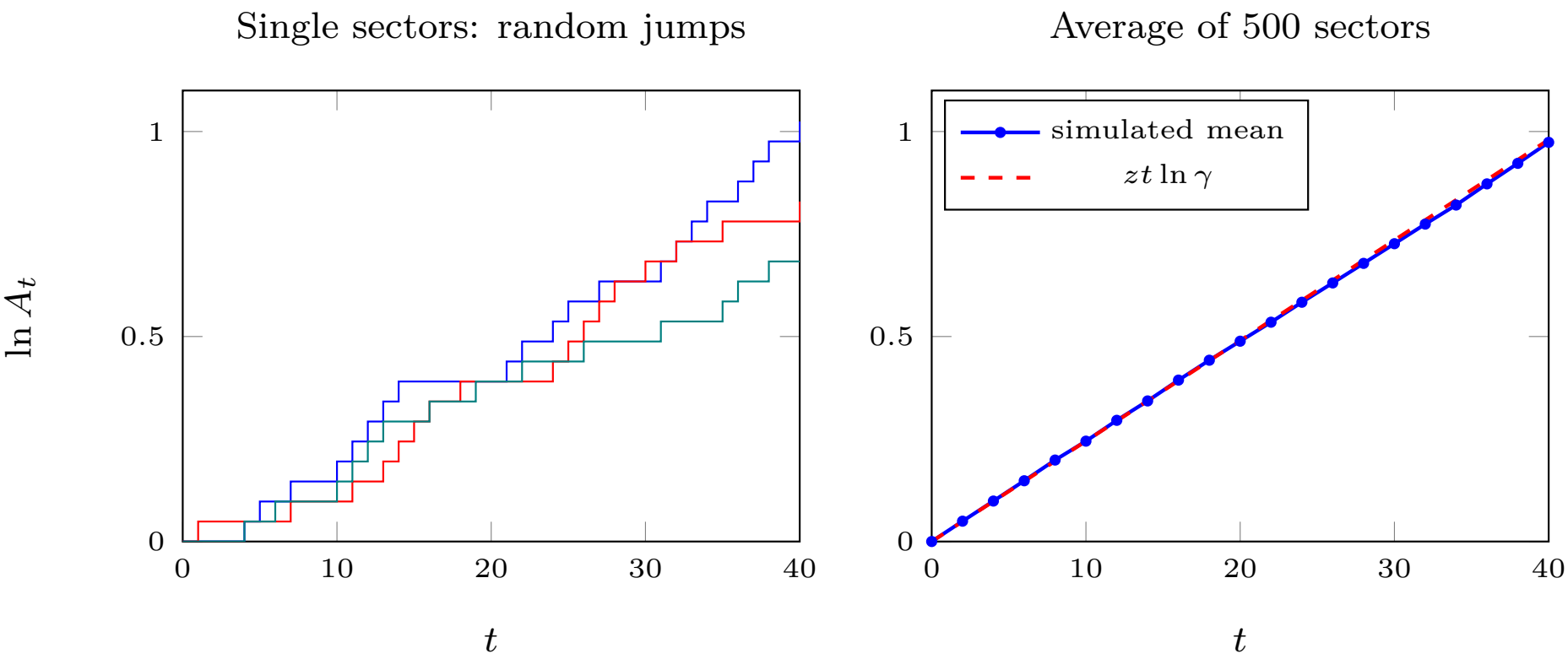
Many sectors: with a continuum of sectors $j \in [0,1]$, independent innovations and $\ln Y_t = \int_0^1 \ln Y_{jt}\, dj$, the law of large numbers implies the fraction of innovating sectors is z:

$$\ln Y_{t+1} - \ln Y_t = z \ln \gamma \quad \text{(deterministic)}.$$

Long-run growth = frequency of innovations $\times$ size of innovations.

Worked example: $g = 0.5024 \times \ln 1.05 \approx 2.45\%$ per period.

Quality Ladders and the Law of Large Numbers



Simulated with the worked-example values $z = 0.5024$, $\gamma = 1.05$. Each sector climbs the quality ladder by random steps of size $\ln \gamma$; the average grows almost exactly at $z \ln \gamma$.

Comparative Statics

Parameter	Effect on z and g	Intuition
$\lambda \uparrow$	$\uparrow$	research more productive
$\gamma \uparrow$	$\uparrow$	bigger prize per innovation
$L \uparrow$	$\uparrow$	larger market (scale effect)
$\rho \uparrow$	$\downarrow$	future rents worth less
$\alpha \uparrow$	$\downarrow$	smaller markup, smaller rents

- For α: $L_x \propto \alpha/[\alpha + \gamma(1-\alpha)]$, whose derivative is $\gamma/[\alpha + \gamma(1-\alpha)]^2 > 0$, so $L_x \uparrow$ and $z \downarrow$.
- More product-market competition (smaller markup $1/\alpha$) reduces innovation in this model.
- As in Romer, there is a scale effect.

Comparing the Models

	Romer (1990)	**Jones (1995)**	**Aghion–Howitt**
Engine of growth	new varieties	new varieties	quality upgrades
Accumulated factor	N	N	A
Growth rate	$\frac{\alpha\theta L-\rho}{1+\alpha+\rho}$	$(1+n)^{\frac{1}{1-\phi}}-1$	$z\ln\gamma$
Scale effect	yes	no	yes
Policy $\rightarrow$ growth	yes	no (levels)	yes

Common thread: growth needs something accumulable with non-diminishing returns — the stock of knowledge in Romer/Jones, the quality index in Schumpeter.

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